Project Evaluation Tools

Project Evaluation Constraints

- Capital Rationing
- Several projects could be taken up together at a time.
- There is no sufficient capital available to support all projects proposed in an organization.
- Mutually Exclusive Projects
- The acceptance of one project precludes the acceptance of other projects
- Replacement Projects
- Selecting one project in place of another existing project or alternate project

Evaluation Tools

- ✓**Payback Period**
- ✓**Accounting Rate of Return**
- ✓**Net Present Value**
- ✓**Internal Rate of Return**
- ✓**Modified Internal Rate of Return**
- ✓**Profitability Index**

Payback Period

- Payback period is the time required to recoup the investment committed to a project.
- Companies following payback period use "stipulated payback period", which acts as a standard for screening the project.
- The payback period is stated in terms of years. This can be stated in terms of percentage also. This is the payback reciprocal rate.
- Payback period (percentage):

1/payback period

Computation of Payback Period

1. When the cash inflows are uniform the formula for payback period is cash outflow divided by annual cash inflow.

$$\text{Payback} = \frac{\text{outflows}}{\text{Annual inflows}}$$

2. When the cash inflows are uneven, the cumulative cash inflows are to be arrived at and then the payback period is calculated through interpolation. Here payback period is the time when cumulative cash inflows are equal to the outflows.

$$\Sigma \text{inflows} = \text{outflows}$$

Decision Rules

A. Capital Rationing Situation

- Select the projects which have payback periods lower than or equivalent to the stipulated payback period;
- Arrange these selected projects in increasing order of their respective payback periods;
- Select those projects from the top of the list till the capital Budget is exhausted.

- **B. No Capital Rationing Situation**

- Select the projects with payback periods lower than or equal to the stipulated period;
- All these projects can be accepted since there is no capital constraint.

- **C. Mutually Exclusive Projects**

- In the case of two mutually exclusive projects, the one with a lower payback period is accepted, when the respective payback periods are less than or equivalent to the stipulated payback period.

Stipulated Payback Period

- Stipulated payback period, broadly, depends on the nature of the business/industry with respect to the product, technology used and speed at which technological changes occur, rate of product obsolescence etc.
- Stipulated payback period is, thus, determined by the management's capacity to evaluate the environment vis-a-vis the company's products, markets and distribution channels and identify the ideal-business design and specify the time target.

Advantages of Payback Period

- It is easy to understand and apply. The concept of recovery is familiar to every decision-maker.
- Companies facing uncertainty - both of product and technology - will benefit by the use of payback period method since the stress in this technique is on early recovery of investment. Companies facing technological obsolescence and product obsolescence - as in electronics/computer industry - prefer payback period method.
- Liquidity requirement requires earlier cash flows. Hence, companies having high liquidity requirement prefer this tool since it involves minimal waiting time for recovery of cash outflows as the emphasis is on early recoupment of investment

Disadvantages of Payback Period

- The time value of money is ignored. For example, in the case of project A, INR 500 received at the end of 2nd and 3rd years are given same weight. Broadly a rupee received in the first year and during any other year within the payback period is given same weight. But it is common knowledge that a rupee received today has higher value than a rupee to be received in future.
  - *But this drawback can be set right by using the discounted payback period method. The discounted payback period method looks at recovery of initial investment after considering the time value of inflows.*
- Another important drawback of the payback period method is that it ignores the cash inflows received beyond the payback period. In its emphasis on early recovery, it often rejects projects offering higher total cash inflow. In the illustration Project B offers total cash inflows of INR 18,000/- but payback period selects Project A only though its total cash inflows are INR 16,000 i.e., INR .2,000 lower than those of project B.
- Investment decision is essentially concerned with a comparison of rate of return promised by a project with the cost of acquiring funds required by that project. Payback period is essentially a time concept; it does not consider the rate of return.

Example

- There are two projects(Project A and B) available for a company, with a life of 6 years each and requiring a capital outlay of INR 9,000/-; and additional working capital of INR 1000/-. The cash inflows comprise of profit after tax (PAT) + Depreciation (DEP) + INTEREST(Tax adjusted) for five years and salvage value of INR 500/- for each project plus working capital released in the 6th year. This company has prescribed a hurdle payback period of 3 years. Which of the two projects should be selected?

Data

	Project A	Project B
Investment	10,000	10,000
Cash inflow	6-years	6-years
Year -1	3,000	1,700
Year -2	3,500	2,500
Year -3	3,500	2,500
Year -4	1,500	2,500
Year -5	1,500	4,000
Year -6	3,000	4,800
	16,000	**18,000**

Project Evaluation

- Payback period for Project A = 3 years
- *(cumulative cash inflows = outflows)*

- Payback period for Project B = 4 years + 800/4000 = 4.2 years = 4 years 2 months and 12 days.

	Project A	Cumulative cash inflows of Project A	Project B	Cumulative cash inflows of Project B
Year -1	3,000	3,000	1,700	1,700
Year -2	3,500	6,500	2,500	4,200
Year -3	3,500	10,000	2,500	6,700
Year -4	1,500	11,000	2,500	9,200
Year -5	1,500	13,000	4,000	13,200
Year -6	3,000	16,000	4,800	18,000

Points to Note

- Interpolation technique is used here to identify the exact period at which cumulative cash inflows will be equal to outflows. The amount required to equate is INR 800, while the returns from the 5^{th} year is 4,000. Hence the addition time duration required to compute the payback period is 800/4000 which is 0.2 years. This can be translated into months as 12 x 0.2 = 2.4 months which is 2 months and 12 days, considering a 30-day month. The interpolation technique is used based on the assumption that cash inflows accrue uniformly throughout the year.
- The decision criteria is to choose between Project A with a payback period of 3 years and Project B with a payback period of 4 years 2 months and 12 days.
- The decision will be Project A will be selected since the payback period is less and is equal to the stipulated payback period.

Discounted Payback Period

	Project A	Discounted Cash Flows	Cumulative cash inflows of Project A	Project B	Discounted Cash Flows	Cumulative cash inflows of Project B
Year -1	3,000	2,727	2,727	1,700	1,545	1,545
Year -2	3,500	2,893	5,620	2,500	2,066	3,612
Year -3	3,500	2,630	8,249	2,500	1,878	5,490
Year -4	1,500	1,025	9,274	2,500	1,708	7,197
Year -5	1,500	931	10,205	4,000	2,484	9,681
Year -6	3,000	1,693	11,899	4,800	2,709	12,391

Assumed discount rate (cost of investment/capital): 10% p.a.
Payback period for Project A = 4 years + 726/931 = 4.78 = 4 Years 9 months and 11 days
(cumulative cash inflows = outflows)
Payback period for Project B = 5 years + 319/2709 = 5.12 years = 5 years 1 month and 43 days.

Accounting Rate of Return

- It is the rate arrived at by expressing the average annual net profit(after tax) as given in the income statement as a percentage of the total investment or average investment.
- The accounting rate of return is based on accounting profits.
- Accounting profits are different from the cash flows from a project and hence, in many instances, accounting rate of return might not be used as a project evaluation decision.
- Accounting rate of return does find a place in organizational decision making when the returns expected are accounting profits and not merely the cash flows.

Computation of Accounting Rate of Return

- 1. The accounting rate of return using total investment.

$$\frac{\text{Net Profit After Tax}}{\text{Total Investment}}$$

- 2.Sometimes average rate of return is calculated by using the following formula:

$$\frac{\text{Net profit after tax}}{\text{Average investment}}$$

- Where average investment = total investment divided by 2

Decision Rules

- **A. "Capital Rationing" Situation**
- 1. Select the projects whose rates of return are higher than the cut-off rate
- 2. Arrange them in the declining order of their rate of return and
- 3. Select projects starting from the top of the list till the capital available is exhausted.
- **B. "No Capital Rationing" Situation**
- Select all projects whose rate of return are higher than the cut-off rate.
- **C. Mutually exclusive projects**
- Select the one that offers highest rate of return.

Advantages Vs. Disadvantages

- It is easy to calculate and
- The percentage return is more familiar to the executives.
- The definition of cash inflows is erroneous; it takes into account profit after tax only. It, therefore, fails to present the true return.
- Definition of investment is ambiguous and fluctuating. The decision could be biased towards a specific project, could use average investment to double the rate of return and thereby multiply the chances of its acceptances.
- Time value of money is not considered here.

Example

- There are two projects (Project A and B) available for a company, with a life of 6 years each and requiring a capital outlay of INR 9,000/- and additional working capital of INR 1000/. The cash inflows comprise of profit after tax (PAT) + depreciation (DEP) + INTEREST(Tax adjusted) for five years and salvage value of INR 500/- for each project at year 6 plus working capital released also in the 6^{th} year.

Data

NET PROFIT AFTER TAX		
Year	**Project A**	**Project B**
1	1,580	280
2	2,080	1,080
3	2,080	1,080
4	80	1,080
5	80	2,580
6	80	1,880
Total Net Profit After Tax	5,980	7,980
Average Annual Net Profit	**5,980/6 = 996.6**	**7,980/6 = 1330**

Project Evaluation

- Considering the working capital released in the 6^{th} year and salvage value of the investment, the total investment will be (10,000-1,500) INR 8500 and the average investment will be (8500/2) INR 4250.
- The rate of return calculations are:
- Net profit after tax as a percentage of total investment
- **Project A** **Project B**

$$\frac{996.6 \times 100}{8500} = 11.7\%$$

$$\frac{1130 \times 100}{8500} = 15.6\%$$

Project Evaluation

- Net profit (after tax) as percentage of average investment
- **Project A** **Project B**

$$\frac{996.6 \times 100}{4250} = 23.4\%$$

$$\frac{1130 \times 100}{4250} = 26.6\%$$

- The decision will be to select Project B since its rate of return is higher than that of Project A if they are mutually exclusive. If they are independent projects both can be accepted if the minimum required rate of return is 11.7% or less.

Net Present Value

- Net present value of an investment/project is the difference between present value of cash inflows and cash outflows. The present values of cash flows are obtained at a discount rate equivalent to the cost of capital.
- Let 'b' be the cash outflow in period 't' where t = 0,1,2,....n
- 'B' be the present value of cash outflows
- 'c' be the cash inflow in period 't'=0,1,2,........n
- 'C' be the present value of cash inflows
- 'K' be the cost of capital

Computation of NPV

$$\text{Present value of cash inflows} = C = \sum_{t=0}^{n} \frac{c_t}{(1+k)^t}$$

$$\text{Present value of cash outflows} = B = \sum_{t=0}^{n} \frac{b_t}{(1+k)^t}$$

$$\text{Net present value (NPV)} = (C - B) \quad \text{or}$$

$$\text{Net present value} = \sum_{t=0}^{n} \frac{(c_t - b_t)}{(1+k)^t}$$

Computation of NPV

- When the cash outflow is required for only one year i.e., in the present year, then the Net present value is calculated as follows:

$$\text{Net present value} = \sum_{t=0}^{n} \frac{c_t}{(1+k)^t} - I$$

- "I" is the initial investment (cash outflow) required by the project.
- The NPV rule uses the project's cash flows (not its accounting earnings) and considers all cash flows during the project's life.

Decision Rule

- **A. "Capital Rationing" situation**
- Select projects whose NPV is positive or equivalent to zero.
- Arrange in the descending order of NPVs.
- Select Projects starting from the list till the capital budget allows.
- **B. "No capital Rationing" Situation**
- Select every project whose NPV >= 0
- **C. Mutually Exclusive Projects**
- Select the one with a higher NPV.

Example

- Assuming that the cost of capital is 6% for our illustration of the project involving a lump sum cash outflow of INR 8,200 and cash inflow of INR 2,000 per annum for 5 years, the Net Present Value calculations are as follows:
- a) Present value of cash outflows INR 8200
- b) Present value of cash inflows
- Present value of an annuity of INR 1 at 6% for 5 years=4.212
- Present value of INR 2000 annuity for 5 years = 4.212 X 2000 = INR 8424
- Net present value = present value of cash inflows - present value of cash outflows= 8424 -8200 = INR 224
- Since the net present value of the project in is positive (INR 224) the Project is accepted.

NPV Advantages

- The NPV rule discounts the cash flows rather than simply totaling them. In other words, in recognizing that INR 1 today is worth more than INR 1 tomorrow, it considers the time value of money.
- The NPV rule considers the scale or magnitude of the initial investment
- The NPV rule is consistent with the firm's goal of maximizing stockholders' wealth.

Internal Rate of Return (IRR)

- The internal rate of return method is also known as the yield method. The IRR of a project/investment is defined as the rate of discount at which the present value of cash inflows and present value of cash outflows are equal.

- This can be restated as the rate of discount, at which the present value of cash flow (inflows and outflows) associated with a project equal zero.

Calculation of IRR

- Let at be the cash flows(inflow or outflow) in period t
- Where t = 0,1,2.......... n years
- Then IRR of the project is found out by solving for the value of 'r' in the following equation:

$$\frac{a_0}{(1+r)^0} + \frac{a_1}{(1+r)^1} + \frac{a_2}{(1+r)^2} + \ldots\ldots + \frac{a_n}{(1+r)^n} = 0$$

$$\sum_{t=0}^{n} \frac{a_t}{(1+r)^t} = 0 \qquad \textbf{Equation -1}$$

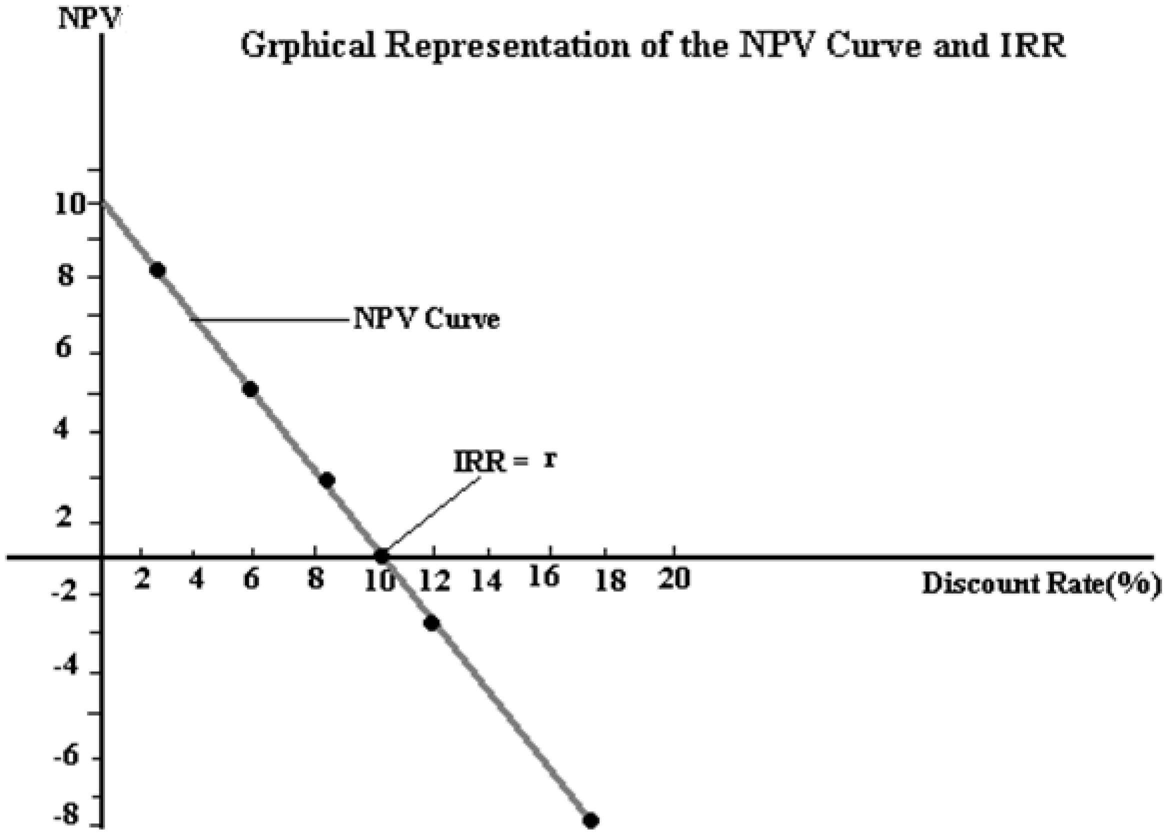
- In the special case where a0 < 0 and at > 0, where t = 1,2,......n

$$a_0 = \sum_{t=0}^{n} \frac{a_t}{(1+r)^t} = 0 \qquad \textbf{Equation -2}$$

Decision Rule

- **A. "Capital Rationing" Situation**
- Select those projects whose IRR (r) = k, where k is the cost of capital.
- Arrange all the projects in the descending order of their Internal Rate of Return
- Select projects from the top till the capital budget allows.
- **B. "No Capital Rationing" Situation**
- Accept every project whose IRR (r) = k, where k is the cost of capital.
- **C. Mutually Exclusive Projects**
- Select the one with higher IRR.

NPV Vs. IRR



NPV
Grphical Representation of the NPV Curve and IRR
10
8
6
4
2
-2
-4
-6
-8
NPV Curve
IRR = r
2
4
6
8
10
12
14
16
18
20
Discount Rate(%)

Example

- A new machinery costs INR 8,200 and generates cash inflow(after tax) per annum of INR 2,000 during its life of 5 years.
- IRR method involves trial and error in the sense that one has to experiment with different rates of discount before arriving at the appropriate rate at which the equation 1 and 2 are satisfied. But when the cash inflows are by way of annuities the relevant interest factor can be found out by the following formula: Cash outflow / Cash inflow per annum
- In the present case this is 8200 divided by 2000 = 4.1
- The interest factor 4.1 for a 5-year project corresponds to a discount rate of 7%. Hence, the IRR of the project is 7%. An interest factor of 4.100 indicates that the present value of one Rupee annuity for 5 years at 7% is equivalent to 4 rupees and ten paise. Hence, the present value of INR 2,000 annuity is 4.100 X 2000 = 8200.
- Present value of cash inflows = INR 8200 and
- Present value of cash outflow = INR 8200
- At 7% the present value of cash inflows is equivalent to the present value of cash outflows.
- Hence, 7% is the IRR of the project.

Trial and Error Method

- Through the trial-and-error method, we can begin with a 10% discount rate. The net present value assuming a 10% discount rate is
- (2000 x 3.7908) - 8200 = 7581.6 - 8200 = -618.4.
- Since the NPV is negative, we need to reduce the discount rate to arrive at a positive NPV. Hence, let us assume a discount rate of 5%. The net present value assuming a 5% discount rate is
- (2000 x 4.3295) - 8200 = 8659 - 8200 = 459.

Trial and Error Method

- Interpolate the IRR which lies between these two positive and negative NPV i.e., the discount rate that results in a 0 NPV.
- For the illustration IRR = 5% +[(618.4/1077.4)x5%] = 5% + 2% = 7%.

$$IRR = r_1 + \left(\frac{N_1}{N_1 + N_2} x (r_2 - r_1) \right)$$

Where

$r_1 =$ smallest rate of interst

$r_2 =$ highest rate of interest

$N_1 =$ NPV at smallest rate

$N_2 =$ NPV at highest rate

Project Evaluation

- If the cost of capital is less than or equal to 7%, we will accept the investment of INR 8200 in the new machine since it offers a return of 7%.
- Such projects whose IRR is equivalent to the cost of capital are called marginal projects since the company on accepting them, will neither be worse off nor better off. They will just pay for themselves.
- Where the cash inflows are not uniform series and/or where cash outflows occur at other periods besides the initial period, it is necessary to experiment different rates and the appropriate rate could be found out through interpolation.

Modified Internal Rate of Return (MIRR)

- Modified internal rate of return is modified to consider the difference between re-investment rate and investment return.

$$MIRR = \sqrt[n]{\frac{Future\ positive\ cash\ inflows}{Present\ value\ of\ cash\ outflows}} - 1$$

Decision Rule

- **A. "Capital Rationing" Situation**
- Select all projects whose MIRR is greater than or equivalent to 1.
- Rank them in the descending order of their MIRR
- Select projects starting from the top of the list till the capital budget amount is exhausted.
- **B. "No Capital Rationing" Situation**
- Select every project whose MIRR >= 1.
- **C. Mutually Exclusive Projects**
- Select the project with higher MIRR.

Example

- A project requires an initial investment of 100,000 and incur net cash outflows of 50,000 for two more years. The cost of capital is 6%. The inflows from the project after two years is expected to give a return of 80,000 for three years.

	Outflow	Inflow	PV Outflow	FV inflow	5^{th} Root (FV/PV)	MIRR	MIRR (%)
0	100000		100000				
1	40000		37735.85				
2	40000		35599.86				
3		80000		89888			
4		80000		84800			
5		80000		80000			
			173335.7	254688	1.080001	0.080001	8.00%

Profitability Index (PI)

- Profitability ratio is otherwise referred to as Benefit/Cost ratio. This is an extension of the Net Present Value Method. This is a relative valuation index and hence is comparable across different types of projects requiring different quantum of initial investments.

$$PI = \frac{\text{Present Value of Cash inflows}}{\text{Present Value of Cash outflows}}$$

- Profitability index is always expressed on net basis. Formula for profitability index(PI) is:
- Where at is the cash inflow in period 't' and t = 1,2,....n
- k0 is cost of capital
- a0 is initial investment.

$$PI = \frac{\sum_{t=0}^{n} \frac{a_t}{(1+Ko)^t}}{a_0}$$

Decision Rule

- **A. "Capital Rationing" Situation**
- Select all projects whose profitability index is greater than or equivalent to 1.
- Rank them in the descending order of their profitability indices
- Select projects starting from the top of the list till the capital budget amount is exhausted.
- **B. "No Capital Rationing" Situation**
- Select every project whose PI >= 1.
- **C. Mutually Exclusive Projects**
- Select the project with higher PI.

Example

- A new machinery costs INR 8,200 and generates cash inflow(after tax) per annum of INR 2,000 during its life of 5 years. Let us assume that the cost of capital for the company is 6%.
- The present value of the cash inflows at 6% discount rate is **2000 * 4.212 = 8424.** The present value of outflows is 8,200. The profitability index is **(8424/8200) = 1.027**.
- The profitability index of **1.027** leads to an acceptance decision of the project, since it is greater than 1.
- So far as "accept-reject" decisions are concerned all the three discounted cash flow (DCF) methods lead to the same decision. But in the case of ranking mutually exclusive projects, sometimes there will be conflicting decisions between NPV and IRR. In these situations, a choice has to be made between these methods. Since PI is a relative ranking method, this fits most suitably for evaluating mutually exclusive projects.

Corporate Preference for Project Evaluation

Project Selection Tool	U.S firms	Non-U.S. firms	Simple average
Internal rate of return	62.0%	34.0%	48.0%
Net present value	9.0%	10.0%	9.5%
Profitability index	3.0%	10.0%	6.5%
Accounting rate of return	14.0%	15.0%	14.5%
Payback period	12.0%	31.0%	21.5%
Total	100.0%	100.0%	100.0%